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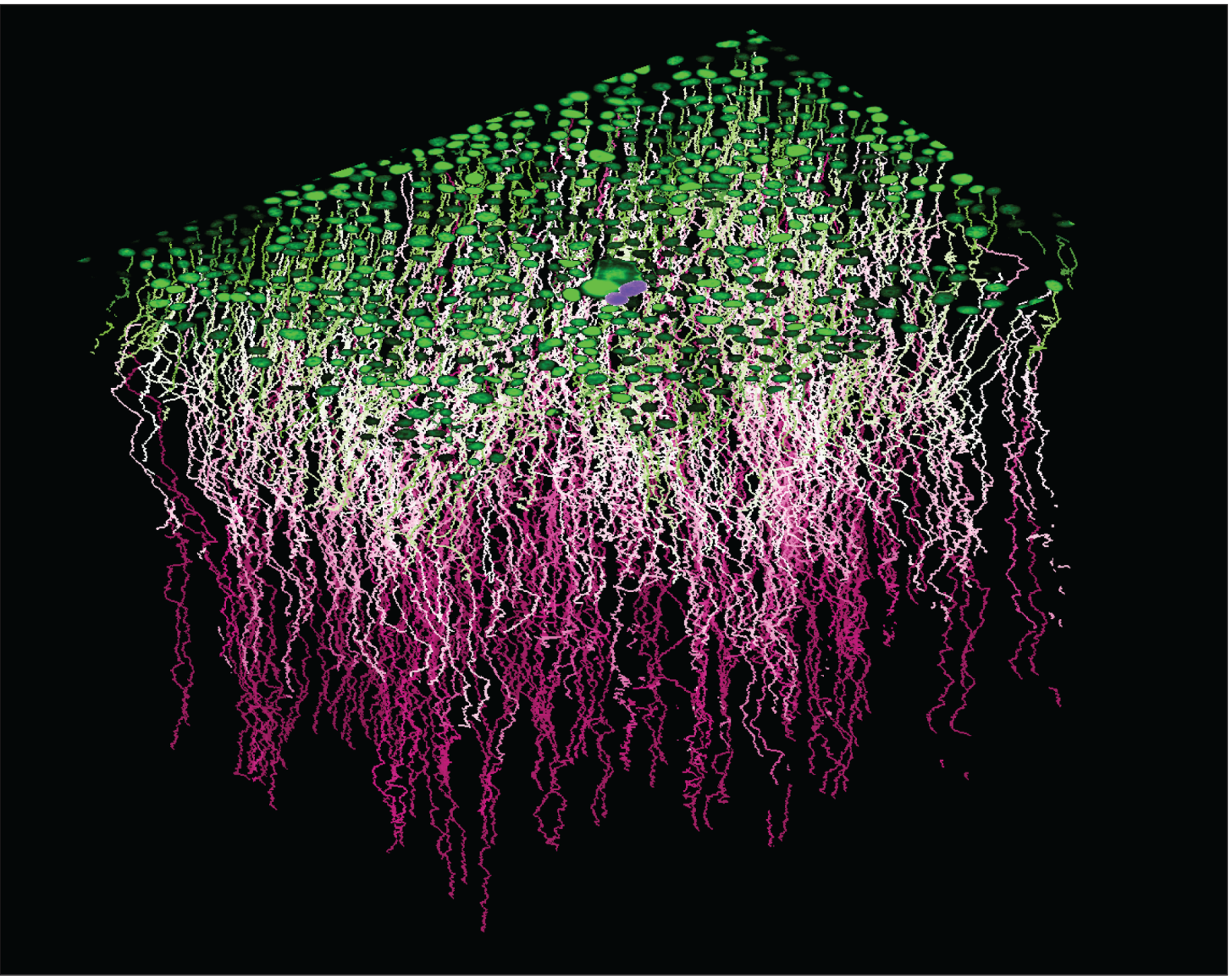
MBoC

MOLECULAR BIOLOGY OF THE CELL



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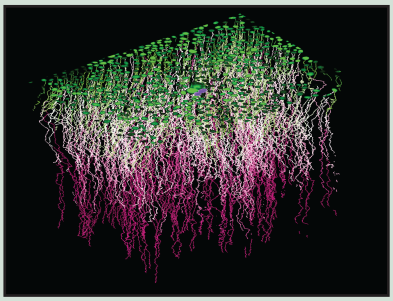

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Tracking the game of life across space and time: A lonely couplet of mutant cells (MDCK RasV12, magenta nuclei) are surrounded by an army of wild-type cells (MDCK, green nuclei) that are conspiring to squeeze out and evict the mutant cells. This is a form of epithelial quality control called cell competition that is governed by organized patterns of spatiotemporally clustered cell divisions, working to push the mutant cells out of the monolayer in which they wrongfully reside. This image shows a 3-dimensional projection of a 5 day time-lapse microscopy acquisition, where the paths that each cell has previously taken are traced out by their tails, an analysis achieved using deep-learning nuclear segmentation and Bayesian multi-object tracking.
All images are attributable to Nathan Day & Jasmine Michalowska under the supervision of Alan Lowe & Guillaume Charras at UCL.

The Philosophy of *Molecular Biology of the Cell*

Molecular Biology of the Cell (*MBoC*) is published by the nonprofit American Society for Cell Biology (ASCB) and is free from commercial oversight and influence. We believe that the reporting of science is an integral part of research itself and that scientific journals should be instruments in which scientists are at the controls. Hence, *MBoC* serves as an instrument of the ASCB membership and as such advocates the interests of both contributors and readers through fair, prompt, and thorough review coupled with responsible editorial adjudication and thoughtful suggestions for revision and clarification. Our most essential review criterion is that the work significantly advances our knowledge and/or provides new concepts or approaches that extend our understanding. At *MBoC*, active working scientists—true peers of the contributors—render every editorial decision.

The Society and *MBoC* are committed to promoting the concept of open access to the scientific literature. *MBoC* seeks to facilitate communication among scientists by

- publishing original papers that include full documentation of Methods and Results, with Introductions and Discussions that frame questions and interpret findings clearly (even for those outside an immediate circle of experts) and
- exploiting technical advances to enable rapid dissemination of articles prior to print publication and transmission and archiving of videos, large datasets, and other materials that enhance understanding.

Scope of *MBoC*

MBoC publishes research articles that present conceptual advances of broad interest and significance within all areas of cell, molecular, and developmental biology. We welcome manuscripts that describe advances with applications across topics including but not limited to: cell growth and division; nuclear and cytoskeletal processes; membrane trafficking and autophagy; organelle biology; quantitative cell biology; physical cell biology and mechanobiology; cell signaling; stem cell biology and development; cancer biology; cellular immunology and microbial pathogenesis; cellular neurobiology; prokaryotic cell biology; and cell biology of disease.

Submissions that report novel methodologies or large datasets are also encouraged, particularly when the technology or data will be widely useful, when it will significantly accelerate progress within the field, or when it reveals a new result of biological significance.

Authors should include with their manuscript submissions all previously unpublished data and methods essential to support the conclusions drawn.